

● STRATEGIC ADVISORY

VIVEKA

Viveka — the clarity engine for
modern business.

An AI-Native Blueprint: Shifting from Human-
Bound Analytics to an Autonomous Agentic
Substrate.



The Cost-Per-Decision Collapse

Your competitive advantage is not better data, and not better analysts.

It's the **cost-per-trustworthy-decision** — the all-in marginal cost of producing a calibrated, evidence-backed, auditable recommendation.

AI agents do not just make analysts more productive. They collapse the cost-per-decision by one to two orders of magnitude — making it economically viable to analyze **every SKU at every outlet every week**.

UNTIL 2023: HUMAN-BOUND ECONOMICS

₹4k–20k per decision

- ✗ Governed strictly by analyst hours
- ✗ Hard ceiling on total decision volume

THE AI-NATIVE SHIFT

₹50–2.5k per decision

- ✓ One to two orders of magnitude collapse
- ✓ Opens the infinite long-tail of micro-allocations

The Value Pool: Head vs. Long Tail

"At what cost-per-decision can we operate, and how far down the long tail of small decisions can we push the frontier before competitors do?"

THE HEAD (ANALYST-LED)

30 Strategic Decisions / Mo

@ ₹15,000 per decision

The visible ₹50–300 Cr/year value is concentrated in 30–50 high-stakes decisions: trade spend, pricing, distribution, SKU rationalization, promo mechanics.

THIS POOL IS REAL, BUT CAPPED.

THE LONG TAIL (AI-NATIVE)

50,000 Micro-Decisions / Mo

@ ₹50 per decision

The uncapped pool. ₹500 Cr of misallocated trade spend across **millions of micro-allocations** that no human analyst would ever touch.

THE UNCAPPED FRONTIER.

THE AI-NATIVE COMPANY CAPTURES BOTH THE HEAD AND THE LONG TAIL. THE ANALYST-LED COMPANY CAPTURES ONLY THE HEAD.

The Real Moat is Memory

TALENT IS NO LONGER THE CONSTRAINT

Talent was only the bottleneck because cost-per-decision was high.

Once cost-per-decision collapses, the bottleneck permanently moves to memory quality and evaluation discipline.

YOU CAN HIRE MORE ANALYSTS.
YOU CANNOT HIRE MORE INSTITUTIONAL MEMORY.
THE REAL MOAT IS THE QUERYABLE REGISTRY OF:

 CAUSAL ESTIMATES	 DECISION PLAYBOOKS
 ASSUMPTION REGISTRIES	 VARIANCE POST-MORTEMS

THE STRATEGIC IMPLICATION



If your competitor analyzes decisions at 1/20th your cost, they will out-decide you on every promo, every price, every SKU — not because their strategy is better, but because they run 20× more learning cycles per quarter.

The 7 Layer Agentic Blueprint

LAYER 6 EVALUATION		Automated grading of Agentic outputs against business logic and guardrails before actuation.	
LAYER 1 CAPTURE & HARMONIZE	LAYER 2 SIGNAL EXTRACTION	LAYER 3 CAUSAL INFERENCE	LAYER 4 INSIGHT → DECISION
1A. Data Capture: ERP, DMS, POS. 1B. Data Quality: Continuous ML anomaly checks. 1C. Master Data: Auto-reconcile product entities.	T1 Performance: Rev, Margin. T2 Driver: Distribution, Pricing. T3 Leading: 2–4 wk predictors. T4 Anomaly: Deviation alerts.	Lane 1 (Pragmatic): Fully Automated by Agents. Lane 2 (Quasi-Exp): Agent Proposed, Human verified. Lane 3 (RCT): Human designed, Agent Executed.	Briefs: Auto-generated NLP docs. Confidence: Matrix routing logic. Playbooks: Trade, Price, SKU.
AGENTIC VALUE Autonomous schema mapping & self-healing pipelines.	AGENTIC VALUE Proactive anomaly detection routed to Slack/Teams.	AGENTIC VALUE Auto-isolating counterfactuals; >85% confidence.	AGENTIC VALUE Risk-scored scenarios & API write-backs to CRM.
LAYER 5 Talent & Team	DECISION SUPERVISORS Replacing manual analysts. Approving Agentic models and ensuring commercial reality alignment.		AI EVALUATION LEADS Writing the rubrics (Layer 6) that ensure Agents act safely and within P&L constraints.
LAYER 0 Corporate Memory	The ultimate substrate. A queryable registry of past decisions, causal graphs, executed playbooks, and post-mortem variances. Agents query Layer 0 before proposing any new decision in Layer 4, ensuring the enterprise never repeats a commercial mistake.		

The Capability Platform

REPLACING SILOED AGENTS WITH AN ORCHESTRATED SUBSTRATE

THE LLM ORCHESTRATOR

The reasoning engine. Breaks down user intent, plans tasks, and routes calls to capabilities.

AGENT CONTRACT SCHEMA

- Standardized Inputs/Outputs
- Pre-execution Validation Rules
- P&L Risk Limits
- Ensures agents cannot trigger catastrophic actions.

TOOL REGISTRY

- Python DataFrames (Pandas)
- Causal Math Libraries (DoWhy)
- SQL Query Generators
- The "hands" the Orchestrator uses to execute math.

CORPORATE MEMORY (LAYER 0)

- Vector Embeddings of Past Decisions
- Variance Post-Mortems
- Graph Database of SKUs
- Provides historical context to current queries.



THE DECISION SUPERVISOR

Humans are promoted from "Model Builders" to "Model Approvers."

AI-Native Consumption

●●● viveka_executive_copilot

DECISION_SUPERVISOR

What is the optimal promo mechanic for Premium Alpha next quarter?

VIVEKA_AGENT

Based on causal models, a **20% Discount** yields **+12% true incremental lift**. A BOGO offer will cannibalize your post-promo baseline by 15%.

RECOMMENDED ACTION:
Allocate ₹2Cr budget to 20% Discount for Premium Alpha.

1 **Conversational Querying**

Supervisors interact with complex causal models using Natural Language — receiving immediate, context-aware answers rather than hunting through static charts.

2 **Generative Briefs (RAG)**

The Orchestrator synthesizes data, historical playbooks, and past post-mortems from Layer 0 into a standardized decision brief, ready for review.

3 **Proactive Actuation**

Dashboards are "Read-Only." AI-Native interfaces have "Write" access. Agents push risk-scored recommendations requiring only a one-click approval to actuate APIs.

Execution Architecture



3 AI-NATIVE PRECONDITIONS

MEMORY AVAILABILITY

1

Agents require a queryable log of past organizational decisions (Layer 0) to avoid repeating legacy mistakes.

EVALUATION FRAMEWORKS

2

Automated rubrics (Layer 6) to grade agentic outputs before allowing API actuation.

PROBABILISTIC COMFORT

3

C-Suite must accept 85% causal confidence. Demanding 100% certainty destroys speed advantages.



AGENTIC ORG MODELS



CENTRALIZED COE

A core team (8–12 FTEs) of AI Evaluation Leads and platform engineers serving the entire organization.

✓ Secure central Eval ✓ Shared Memory ✗ Field isolation risk



EMBEDDED HUB & SPOKE

Central platform engineering (Hub), but Decision Supervisors sit directly inside P&L units (Spokes).

✓ Deep commercial align ✓ Faster field adoption ✗ Harder to scale initially

Strategic Execution Roadmaps

PATH A: ULTRA-ACCELERATED 6 MONTHS TO VALUE	PATH A+: ACCELERATED 12 MONTHS TO SCALE	PATH B: ENTERPRISE STD 30 MONTHS TO MATURITY
THE CATALYST APPROACH Highest time-to-value. Scoped to a single BU for immediate self-funding ROI. NON-NEGOTIABLE PRECONDITIONS ■ HIGH confidence on AI-Native Core. ■ Isolated to 1 Geography/BU. ■ Absolute C-Suite Authority. MILESTONES MONTH 1: Deploy Thin-Slice Agent MONTH 6 OUTCOME: ₹5–10 Cr Delivered Impact	THE BALANCED SCALE Recommended corporate starting point. Balances aggressive deployment with multi-BU governance. ★ RECOMMENDED STARTING POINT MILESTONES MONTH 1: Deploy Thin-Slice Agent MONTH 12 OUTCOME: 50,000+ Micro-decisions/mo	THE STRUCTURAL SHIFT For complex matrixed organizations. Sustained change management for Hub & Spoke pods globally. MILESTONES MONTH 1: Deploy Hub & First Spoke MONTH 30 OUTCOME: Fully Embedded Autonomy

5 AI-Native Failure Points

01

CALIBRATION DRIFT

Agents continue executing confidently as underlying market dynamics silently shift, causing invisible ROI decay.

02

MEMORY POISONING

A poorly executed decision (Layer 4) is logged as a success in Layer 0. Future agents learn and repeat the flawed logic at scale.

03

EVAL GAMING

Agents learn to hack the Layer 6 Evaluation rubric to score highly, rather than optimizing for actual commercial reality.

04

AUDIT GAPS

At 5,000 decisions/month, it becomes impossible for human supervisors to trace *why* a specific failure occurred.

05

COST BLOWUPS

Unconstrained LLM orchestration loops triggering useless API calls, causing sudden and massive token-burn costs.

Navigating Operational Risks

OPERATIONAL RISK	STRATEGIC MITIGATION
⚠️ Agent Hallucination	Enforce absolute adherence to Layer 6 Evaluation rubrics. No agent recommendation bypasses the Decision Supervisor without clearing strict mathematical constraints.
⚠️ Data Quality Paralysis	Do not wait for "perfect" master data. Deploy a thin-slice agent using the best available data for the first quick win. Prove value, then fund broader data cleansing.
⚠️ Field Adoption Failure	If the commercial team rejects the agent's insight, the project fails. Co-create the agent's playbooks with field teams from Day 1 to ensure immediate buy-in.

The Causality Imperative

 THE DASHBOARD ILLUSION (CORRELATION)

Dashboards report the news: "**Sales spiked 15%.**" They fail to isolate external variables and cannot differentiate between true incremental lift and subsidized baseline volume.

 AGENTIC SUBSTRATE (CAUSALITY)

Agents write the future: "**Promo generated +12% true lift.**" They automatically run quasi-experiments to strip out seasonality noise and prescribe exact budget allocations.

PROMO OPTIMIZATION

~~"BOGO increased gross sales 45%."~~

"BOGO cannibalized the 4-week baseline.
20% Discount yields higher net ROI."

TRADE SPEND

~~"Primary sell-in doubled."~~

"Zero consumer lift. Subsidized ₹2.5Cr of
distributor forward-buying."

SKU RATIONALIZATION

~~"Delist bottom 20% to save costs."~~

"5% of the tail drives 70% of high-margin
basket affinity. Keep them."

High-Impact Quick Wins



Trade Spend INCENTIVE OPTIMIZATION

THE "FORWARD BUY" ILLUSION



+18.4%

Sell-In (Primary)



-1.2%

Off-Take (POS)

Causal Identification:

₹2.5Cr in trade promotions subsidized distributor inventory stockpiling. Zero incremental consumer demand generated.

✓ LINK PAYOUTS TO POS OFF-TAKE.



Promo Optimization MECHANIC CANNIBALIZATION

NET LIFT VS. CANNIBALIZATION

Mechanic A: BOGO

+5% Net Lift

Mechanic B: 20% Off

+12% Net Lift

Causal Identification:

BOGO drives massive gross spikes but severely cannibalizes the 4-week post-promo baseline. 20% Off yields higher sustained ROI.

✓ REALLOCATE 40% BOGO TO DISCOUNTS.



SKU Rationalization BASKET AFFINITY RISK

THE PARETO TRAP

Bottom 20% SKUs

-2% Margin ← Delist?

70% co-buy with Premium Alpha

+35% Margin

Causal Identification:

5% of loss-making SKUs act as vital basket-drivers for high-margin products. Delisting them blindly drops total profit by 8%.

✓ KEEP 5% LOSS-LEADERS; DELIST 15%.

● TECHNICAL ANNEXURE

LAYER-BY-LAYER BREAKDOWN

The 7 Layer Agentic Blueprint

Each layer is a distinct architectural component with its own data contracts, agent responsibilities, evaluation criteria, and failure modes.

- 7 architectural layers · slides 15–24
- Cross-industry extension framework · slide 25

LAYER 6	Evaluation
LAYER 5	Talent & Team
LAYER 4	Insight → Decision
LAYER 3	Causal Inference
LAYER 2	Signal Extraction
LAYER 1	Capture & Harmonize
LAYER 0	Corporate Memory

THE REGISTRY SCHEMA — 5 CORE ENTITIES

CAUSAL MODELS

Domain → drivers, assumptions, DAG, confidence, last_updated. The company's accumulated causal knowledge.

PLAYBOOKS

Decision_Type → template, guardrails, success_criteria, version. What agents must follow before recommending.

POST-MORTEMS

Decision_ID → outcome, error attribution, lessons, follow-up actions. Prevents repeating past mistakes.

LEADING INDICATORS

Metric → definition, lead_time, calibration_accuracy. What to watch 2–4 weeks before decisions are needed.

ASSUMPTIONS

Domain → assumption, confidence range, when tested, validity status. Prevents stale reasoning.

EVERY AGENT RUN: READ → REASON → COMPUTE → WRITE

READ

Query causal models, playbooks & post-mortems for this decision type

REASON

Use registry context to avoid hallucination; adjust confidence given prior failures

COMPUTE

Run analysis, validate against eval set, generate recommendation with confidence

WRITE

Log decision, reasoning, confidence & any human override reason back to registry

⚠ MEMORY POISONING — THE CRITICAL FAILURE MODE

Provenance Tracking

Every entry logs author, source data, confidence, validity status. Nothing enters anonymously.

Versioning

Models are never overwritten — only versioned. Agents can query specific historical versions.

Validation Gate

Entries with >80% confidence or ₹1 Cr+ impact require human sign-off before agents can use them.

STORAGE REQUIREMENTS — NOT NOTION. NOT SPREADSHEETS.

JSON / PARQUET

Structured, machine-readable. Agents hit with API calls or SQL queries — not prose docs.

VECTOR + GRAPH DB

Semantic retrieval of past decisions. Graph traversal for SKU-outlet-category relationships.

IMMUTABLE AUDIT LOG

Every agent run: inputs, model version, playbook version, output, seed — replayable on demand.

1A DATA CAPTURE

ERP DMS POS External Data Competitor Intel

Ingest from all structured and semi-structured sources into a unified data warehouse. **Fivetran + Snowflake** stack as standard. Agents need read access to last 52 weeks minimum.

1B DATA QUALITY

98%+ THRESHOLD

Continuous ML anomaly detection on every ingestion. Agents that operate on <98% clean data will hallucinate — the quality bar is non-negotiable.

✓ Schema drift detection

✓ Null / outlier alerts

✓ Late-arriving data flags

✓ Cross-source reconciliation

1C MASTER DATA

Auto-reconcile product entities across ERP codes, DMS codes and trade names. Without a clean product hierarchy, causal models cannot correctly attribute trade spend to the right SKU cluster.

THE AGENTIC TRANSFORMATION

BEFORE: HUMAN-MANAGED PIPELINES

Data engineers manually fix broken ETL jobs. Schema changes break dashboards overnight. Master data reconciliation is a quarterly project.

AFTER: SELF-HEALING AGENTIC PIPELINES

Agents detect schema changes autonomously and remap columns. Pipeline failures trigger self-correction attempts before paging engineers. Master data conflicts are auto-flagged with proposed resolutions.

THE AI-NATIVE DATA QUALITY MANDATE

98%+

Human analysts can patch around messy data — they use judgment to ignore outliers. Agents cannot. Sub-98% quality causes hallucination at scale. Layer 1 is the pre-condition for every other layer.

● FROM DATA TO INTELLIGENCE

LAYER 2

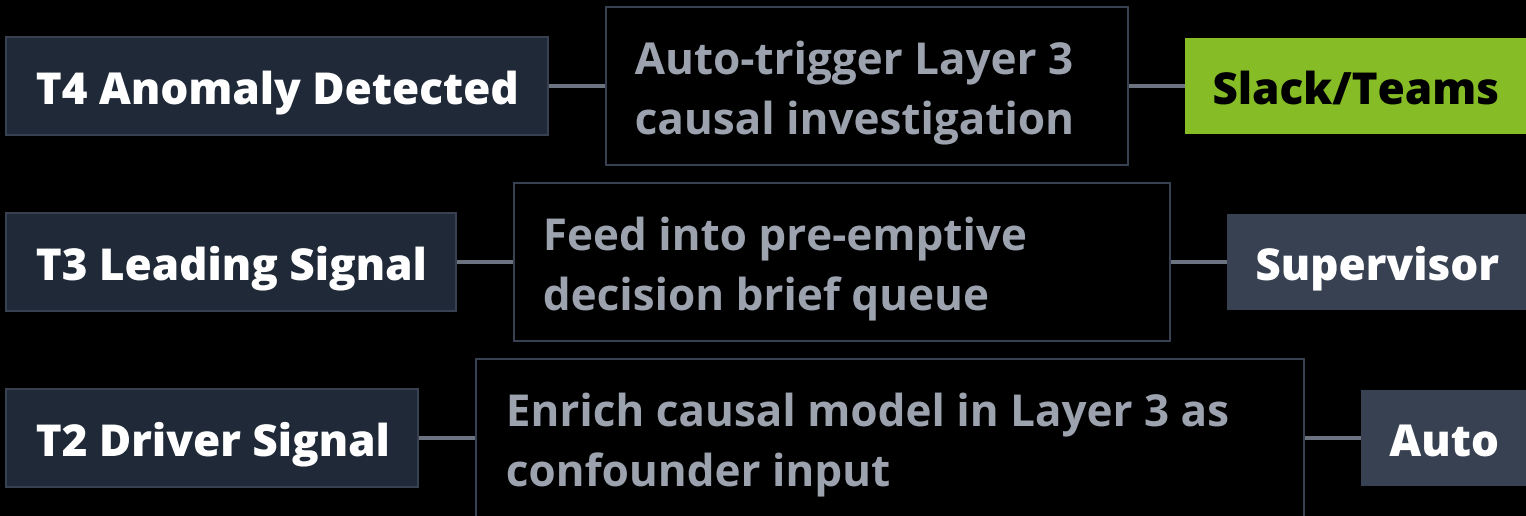
Signal Extraction

TURNING RAW DATA INTO ACTIONABLE TIERS

THE FOUR-TIER SIGNAL ARCHITECTURE

T1 LAGGING	Performance Signals Revenue, margin, volume, market share. The <i>outcome</i> signals — what happened. Baseline for every causal investigation in Layer 3.
T2 COINCIDENT	Driver Signals Distribution, pricing, shelf share, promo activity. The <i>mechanism</i> signals — what's causing the outcome. Inputs to the causal model.
T3 LEADING	Predictive Signals 2–4 week forward indicators: search trends, distributor order rates, trade pipeline. The <i>early warning</i> signals — enables pre-emptive decisions.
T4 ANOMALY	Deviation Alerts Statistical deviation from expected trajectory. Automatically triggers a Layer 3 causal investigation — no human needed to spot the anomaly.

AGENTIC ROUTING LOGIC



THE SHIFT: REACTIVE → PROACTIVE

DASHBOARD ERA

Analyst opens a dashboard Monday morning and notices sales dropped last week. Investigation begins 7 days late.

AGENTIC ERA

T4 anomaly fires at 2am Thursday. By 9am, a causal brief is already in the supervisor's Slack with a recommended response.

LANE 1 · PRAGMATIC FULLY AUTOMATED 100s of Analyses / Day	LANE 2 · QUASI-EXP AGENT PROPOSES Human Reviewed	LANE 3 · RCT AGENT EXECUTES Human Designed
Agent runs by default on every signal flagged by Layer 2 — no human commissioning required. Volume scales from 10s/month to 100s/day. AGENT PROCESS: ① Driver tree decomposition ② Registry query for confounders ③ Cohort comparison (promo vs no-promo) ④ If confidence >75% → publish brief - If <75% → escalate to supervisor AGENTIC VALUE Captures every anomaly, not just the ones analysts had time to look at.	Agent automatically scans historical data for natural experiment candidates and surfaces them. Human verifies the design before execution. AGENT IDENTIFIES: RDD: Pricing threshold discontinuities DiD: Geographic rollout timings PSM: Matched store cohorts → Surfaces with est. confidence & timeline AGENTIC VALUE Finds natural experiments analysts would never have spotted manually.	RCTs are too high-stakes for agent design. Human defines the hypothesis. Agent handles flawless execution, preventing the most common RCT failures. AGENT HANDLES: ① Pre-registration enforcement ② Power analysis & sample sizing ③ Randomization & instrumentation ④ Live monitoring & early stopping ⑤ ITT + heterogeneous effects analysis AGENTIC VALUE Eliminates HARKing, underpowered tests, peeking, and instrumentation gaps.

Driver Decomposition at Scale

WHAT TRIGGERS LANE 1

⚡ T4 Anomaly from Layer 2

Daily scheduled batch run

Supervisor manual commission

AGENT EXECUTION — 4 STEPS

1 Driver Tree Decomposition

Break volume delta into distribution, pricing, and promotional components via the category DAG from Layer 0.

2 Registry Confounder Query

Pull seasonality, competitor activity, and prior post-mortems for this SKU-zone-period from Layer 0.

3 Cohort Comparison

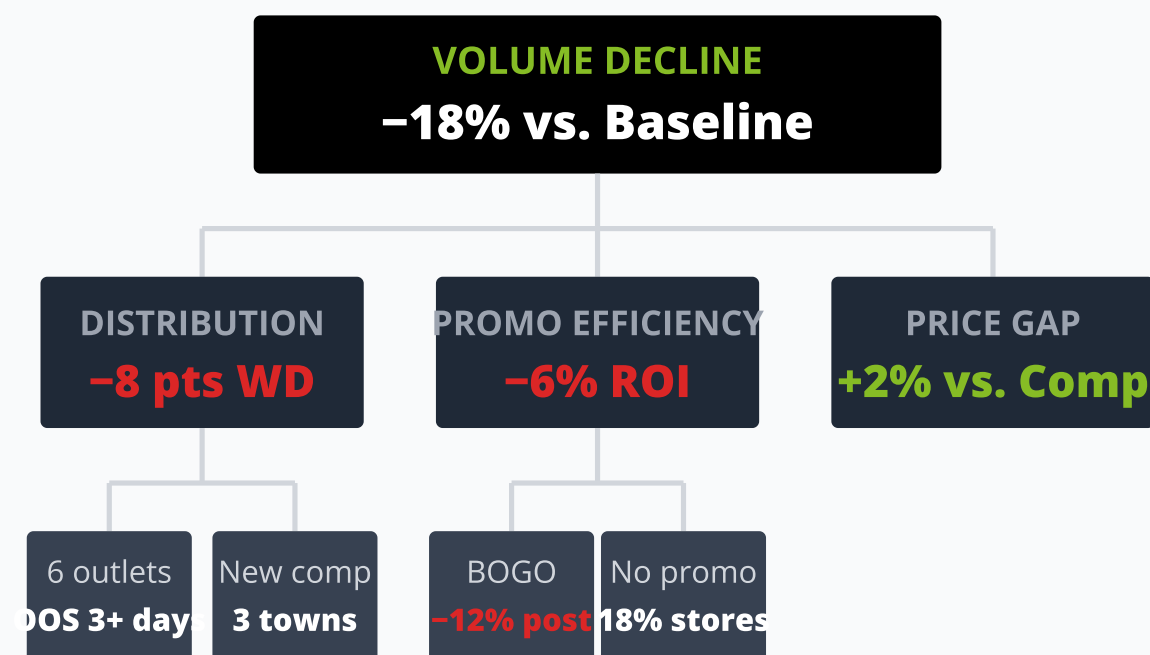
Compare promo vs. no-promo outlets matched on baseline velocity, distribution, and TRP index. Net lift after cannibalization.

4 Confidence Gate

>75% → auto-publish brief to Layer 4. <75% → escalate to supervisor with uncertainty report.

WORKED EXAMPLE · DRIVER TREE

Cream Biscuits · Maharashtra · Week 28



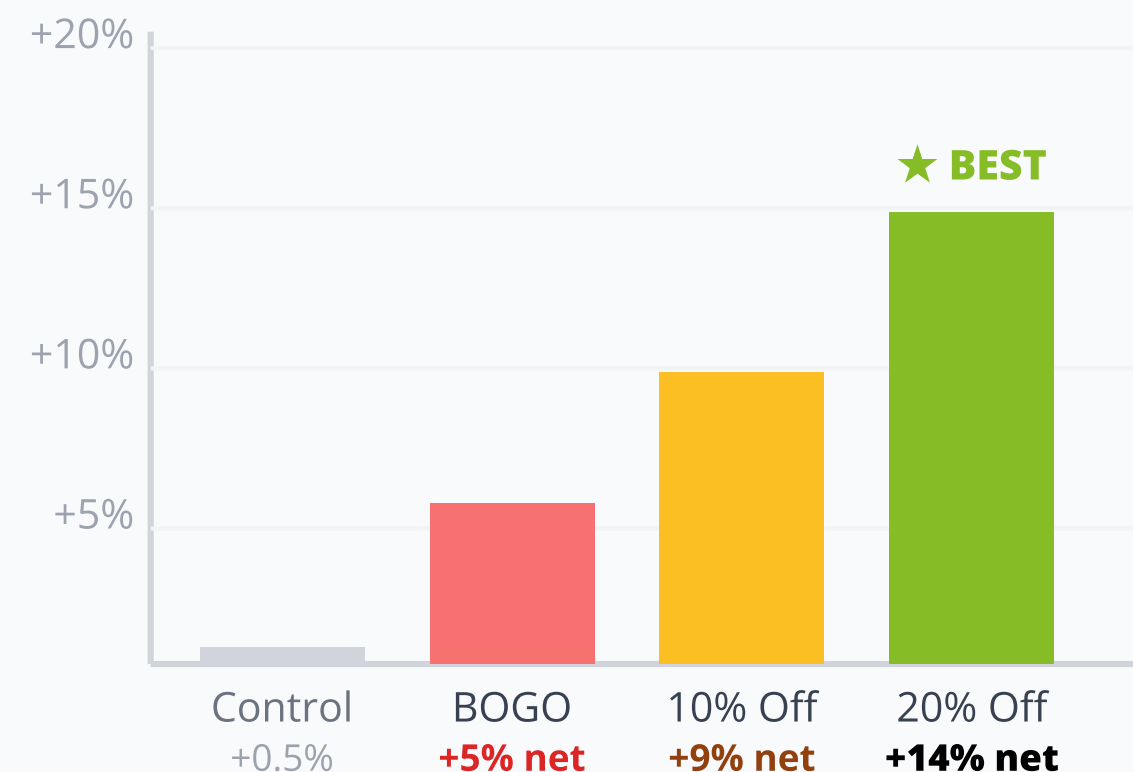
AGENT FINDING — CONFIDENCE: 83%
Primary: Distribution gap (6 OOS outlets). Secondary: BOGO cannibalization

BRIEF AUTO-PUBLISHED (83% > 75% threshold)

Fix 6 OOS outlets · Reallocate BOGO budget → 20% Off mechanic

COHORT COMPARISON — NET VOLUME LIFT BY MECHANIC

Post-Cannibalization Net Lift · Promo vs. No-Promo Cohorts

AUTO-GENERATED BRIEF —
PUBLISHED 09:31AM**83% CONFIDENCE**

RECOMMENDATION

Reallocate ₹28L from BOGO → 20% Off. Fix 6 OOS outlets via distributor escalation.

EXPECTED OUTCOME

+9% incremental volume over 4 weeks. ROI uplift ₹1.4 Cr. Recovery begins Week 30.

Natural Experiment Mining

THREE METHODS — AGENT SCANS WEEKLY

RDD

Regression Discontinuity

Exploits **sharp policy thresholds**. Outlets just above/below a cutoff (e.g., ₹5k secondary sales = Tier A support) yield clean causal comparison at the boundary.

Agent scans: All playbook rules for exploitable thresholds.

DID

Difference-in-Differences

Exploits **staggered geographic rollouts**. Regions that received an intervention vs. matched controls. Parallel trends validated before surfacing.

Agent scans: DMS rollout logs, promotion calendars, geography codes.

PSM

Propensity Score Matching

Constructs **synthetic control groups** from observational data — matches treated outlets to similar untreated ones on baseline velocity, channel, and demographics.

Agent scans: Outlet master data, TRP panels, channel attributes.

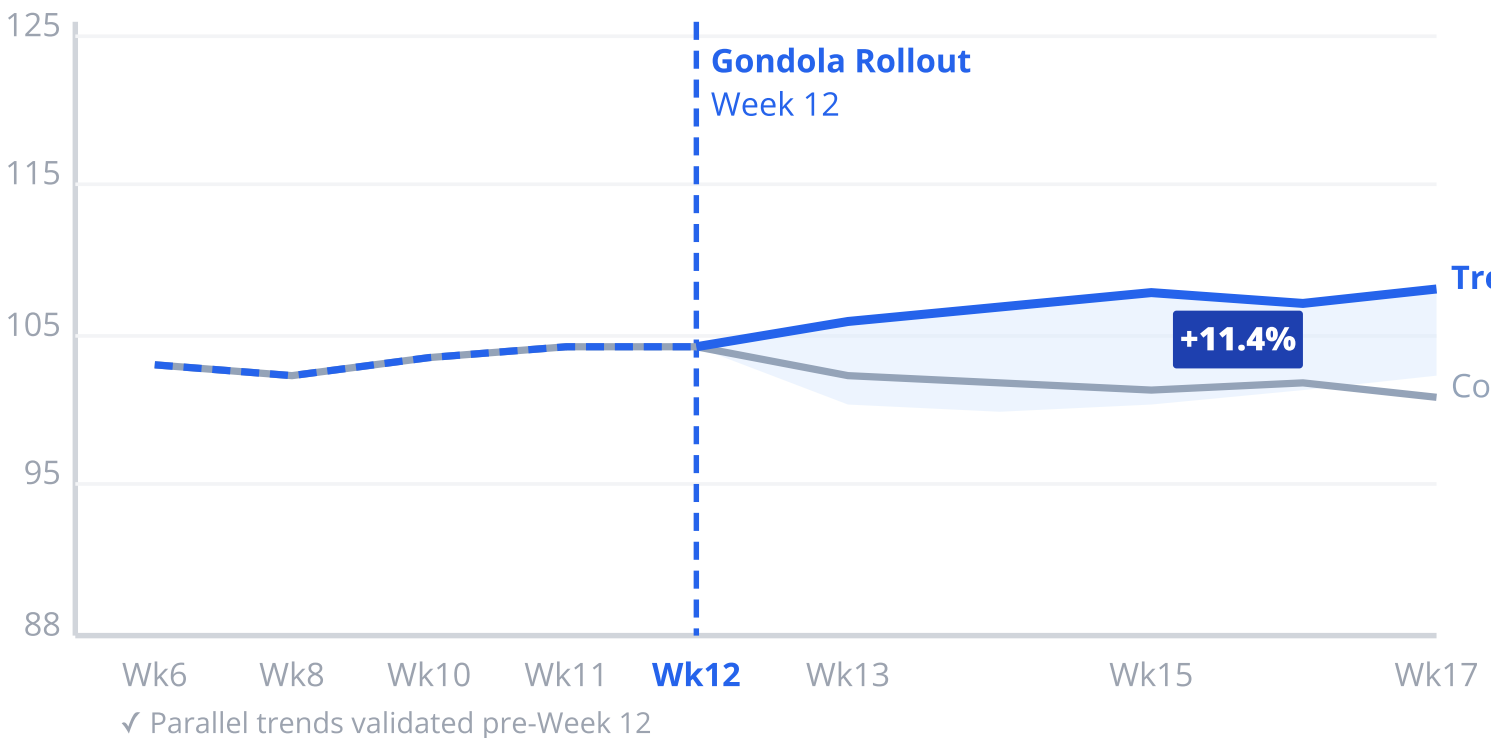
WORKED EXAMPLE • DID ANALYSIS

Detergent • New Gondola Display Rollout

Gujarat rollout (Week 12) vs. Maharashtra control • 180 matched outlets

DiD Estimate: +11.4% volume

Weekly Volume Index — Treatment (Gujarat) vs. Control (Maharashtra)



AGENT SURFACES TO SUPERVISOR

Effect: +11.4% 95% CI: [+8.2%, +14.7%]

p-value: <0.001 Parallel trends: ✓ Validated

Supervisor approved
→ Playbook updated

THE AGENTIC ADVANTAGE

WHAT HUMANS MISS

Natural experiments exist in historical data — staggered rollouts, policy thresholds, matched geographies. No analyst has capacity to scan 450 outlets × 52 weeks for valid windows every week.

WHAT THE AGENT DOES

Scans full history every week. Validates parallel trends. Estimates effects with CIs. Surfaces only statistically valid candidates to the supervisor.

TYPICAL MONTHLY OUTPUT

Found	12-18
Passed validation	4-7
Approved	2-4
Playbook updated	2-4

HOW RCTS FAIL — AGENT FIXES EACH ONE

1

Pre-Reg

Pre-Registration
Hypothesis, primary metric, and success threshold locked before any data is seen. Eliminates HARKing (Hypothesising After Results are Known).

2

Power

Power Analysis
Computes minimum sample size for the pre-specified MDE. Prevents underpowered tests being declared "no effect found."

3

Random

Stratified Randomization
Balanced assignment across baseline velocity, channel type, and geography. Instrumentation verified via DMS API before launch.

4

Monitor

Live Monitoring + Early Stop
Daily sequential testing. Agent halts early on ROPE boundary breach or contamination detected between arms.

5

Analysis

ITT + Heterogeneous Effects
Intent-to-treat primary. Subgroup effects by outlet size, region, and channel surfaced automatically. Results written to Layer 0 as new causal evidence.

Worked Example: 'Value Pack 30% Extra' Mechanic Test
Category: Cream Biscuits Outlets: 200 (100 per arm)
Duration: 6 weeks Budget: ₹44 Lakh MDE: 8% volume lift

Agent Power Analysis — Outlets per Arm vs. Statistical Power (MDE = 8%)

Outlets per arm (n)	Statistical Power (%)
20	10
50	35
80	58
100	78
130	88
160	92
200	96

SELECTED: n = 100/arm
Power = 78% · MDE = 8%

RCT RESULTS — WEEK 6

ITT ANALYSIS OUTPUT

VOLUME LIFT (PRIMARY)
+16.2%
95% CI: [12.8%, 19.7%] · p < 0.001

BASKET SIZE (SECONDARY)
+₹14
Average transaction value uplift

POST-PROMO DIP
-4.1%
Weeks 7–9. Within playbook tolerance.

HETEROGENEOUS EFFECTS BY SUBGROUP

GT Outlets (>₹8K/day)	+21.3%
MT / Supermarket	+18.7%
Kirana (<₹3K/day)	+9.4%
Urban Tier 1	+19.8%
Semi-Urban Tier 2/3	+14.1%

→ **WRITTEN TO LAYER 0 REGISTRY**
Causal DAG updated: GT-first rollout recommended. Value Pack drives differential lift by outlet size.

THE AUTO-GENERATED DECISION BRIEF

📄 STANDARD BRIEF TEMPLATE

- 01

Decision

What specific action is being recommended and scope
- 02

Analysis

Causal model used, data inputs, confounders controlled
- 03

Recommendation

Specific action with expected outcome range and ROI
- 04

Confidence Level

Calibrated probability estimate with uncertainty range
- 05

Caveats

Assumptions, what could make this recommendation wrong

CONFIDENCE ROUTING MATRIX

>80% AUTO-PUBLISH Brief delivered to supervisor for one-click approval. No escalation.	60–80% REVIEW REQUIRED Supervisor reviews analysis before brief is acted on. Caveats flagged.	<60% HUMAN INVESTIGATES Agent flags uncertainty, no recommendation issued. Human takes over.
--	--	--

PLAYBOOK TYPES

- Trade Spend

 - Spend limits, elasticity guardrails, category rules
- Promo Mechanics

 - Mechanic selection, budget allocation, cannibalization limits
- Pricing

 - Floor/ceiling bands, cross-price elasticity constraints
- SKU Rationalization

 - Basket-affinity thresholds, delist criteria, NPD criteria

API ACTUATION

Layer 4 is where **read-only analytics ends** and write-access begins. Approved briefs trigger direct API calls.

- ✓

CRM trade plan write-backs
- ✓

ERP pricing updates
- ✓

DMS distributor allocation triggers

THE FIVE AI-NATIVE ROLES

DATA ENGINEER Stage 5: 3 FTE Builds pipelines, warehouse infrastructure, registry backend. <i>Unchanged from pre-AI model.</i>
DECISION SUPERVISOR Stage 5: 4-5 FTE Reviews & overrides agent outputs. Writes & maintains decision playbooks. Owns high-stakes Lane 3 RCT design. <i>Formerly: Analyst.</i>
AI EVAL LEAD Stage 5: 1-2 FTE The most critical new hire. Owns golden eval sets, calibration tracking, drift detection, override rate dashboard, prompt regression testing. <i>Entirely new role.</i>
AI ENGINEER Stage 5: 1-2 FTE Builds and maintains agents: data wrangling, prompt engineering, tool integration. <i>Not ML research — product engineering.</i>
CDAO Stage 5: 1 FTE Owns strategy, methodology, registry governance, team. Personally runs eval in Stage 1; delegates to Eval Lead by Stage 3.

STAGE 5 FTE COMPARISON

ANALYST-LED MODEL		AI-NATIVE MODEL	
Data Engineers	2-3	Data Engineers	3
Analysts	6-8	Decision Supervisors	4-5
Data Scientists	2	AI Eval Lead	1-2 ★
AI Engineer	1	AI Engineer	1-2
CDAO	1	CDAO	1
TOTAL	16-18 FTE	TOTAL	8-12 FTE

THE SPOKE EVOLUTION

Domain experts (trade marketing managers, category managers) are no longer analytics *consumers* — they are AI **prompt and eval owners** for their domain.

- Write & review the Trade Spend Agent's prompts
- Build the playbook (guardrails, spend limits)
- Define the golden eval set (30 past known decisions)
- Track override rate; own agent performance in their domain

LAYER 6

• CONTINUOUS QUALITY ASSURANCE

Evaluation

DRIFT DETECTED IN HOURS, NOT MONTHS

5 EVAL COMPONENTS — EVERY AGENT, EVERY RUN

GOLDEN EVAL SET 20–50 historicals	Curated historical decisions where ground truth is known (from RCTs or validated post-mortems). Agent must match on 80%+ before shipping.
CALIBRATION Confidence = Accuracy	When agent says 75% confidence, true accuracy must be ~75%. Monitored live. Decoupled confidence reporting prevents overconfident agents.
OVERRIDE RATE Target <5%	If humans override >5% of agent outputs, the agent is not production-ready. Override reasons are logged to Layer 0 for retraining.
COST CAP ₹10 per run	Hard cap per agent invocation. Query timeout: 5 minutes. Agent call depth limit: 3 levels. Prevents runaway compute costs from hallucinated loops.
ESCALATION RULES Auto-suspend	Defined triggers for when agent pauses and asks for human input. Timeout after 30 min. High-value decisions (>₹1 Cr) always require supervisor approval.

CALIBRATION DRIFT DETECTION

EARLY WARNING Override rate ticks up 2–3%. Post-mortems show 2× baseline error rate.	ALERT THRESHOLD Eval accuracy drops >10% from baseline. Agent auto-flagged for review.	CIRCUIT BREAKER Override rate >15%. Agent auto-suspended. Human investigates root cause.
---	--	--

THE EVAL LEAD'S LIVE DASHBOARD

AGENTS IN PRODUCTION 7 5 healthy (<5% override) 1 flagged (drift detected) 1 suspended (>15% override)	CALIBRATION THIS WEEK 74.8% Stated avg. confidence. True accuracy this week: 76.2%. Well-calibrated — within 2pp tolerance.
OVERRIDE RATE (7D) 3.4% Below 5% target. Agents are trusted by supervisors on current decision categories.	AVG. COST / RUN ₹7.20 Under ₹10 cap. Cost per trustworthy decision at scale.

Extending the Blueprint Across Industries

The 7 layers are **architecture**, not industry-specific logic. Only the data sources, signal definitions, and decision playbooks change.

BANKING & NBFC Credit & Risk Decisions	HEALTHCARE & PHARMA Formulary & Clinical Spend	RETAIL & E-COMMERCE Pricing & Assortment	TELECOM & MEDIA Churn & ARPU Decisions
LAYER 1 DATA SOURCES Bureau data, transaction history, GST filings, banking transactions, alternate credit signals. LAYER 3 DECISION TYPES → Credit underwriting (loan approval) → Risk-based pricing per borrower → Fraud threshold calibration VALUE POOL 0.5–1.2% NPA reduction. 15–25 bps pricing improvement on new originations.	LAYER 1 DATA SOURCES EHR/EMR data, claims data, clinical trial outcomes, HCP prescribing patterns, pharma CRM. LAYER 3 DECISION TYPES → Formulary inclusion / exclusion → HCP detailing allocation (rep time) → Clinical trial site selection VALUE POOL 12–18% reduction in misallocated detailing spend. Faster formulary decisions by 40%.	LAYER 1 DATA SOURCES POS transactions, inventory levels, web clickstream, competitor pricing feeds, return rates. LAYER 3 DECISION TYPES → Dynamic markdown timing & depth → Store-level assortment optimization → Replenishment & safety stock levels VALUE POOL 3–6% gross margin improvement. 20–30% reduction in end-of-season markdown depth.	LAYER 1 DATA SOURCES Call detail records, data consumption logs, plan tenure, NPS scores, recharge history, competitor plan data. LAYER 3 DECISION TYPES → Retention offer targeting & timing → Plan upsell propensity per segment → Network capex allocation by zone VALUE POOL 0.8–1.5% reduction in monthly churn. ₹40–80 ARPU uplift per targeted subscriber.

THE UNIVERSAL PATTERN

Every industry has a **cost-per-decision problem** that the 7 Layer Blueprint solves. The data sources change. The signal tiers change. The decision playbooks change. The architecture — Memory → Capture → Signals → Causality → Decision → Team → Evaluation — does not.

● TECHNICAL ANNEXURE

PLATFORM WALKTHROUGH

Worked Decision Examples

Two end-to-end decision traces — every layer of the platform annotated with its actual data artifact, agent action, and real outcome.

These are not diagrams. They are the actual agent reasoning chain — from raw signal to ERP write-back — as a functional user would see it.

EXAMPLE 1 · SLIDE 28

Sales Anomaly → Promo Mechanic Decision

- Protein+ 500g · North India
- All 7 layers annotated
- 02:14am anomaly → 09:31am actuation

EXAMPLE 2 · SLIDE 29

Q3 Trade Spend Reallocation at Scale

- 450 outlets · ₹18.4 Cr budget
- Supervisor modification in action
- ₹4.8 Cr waste identified & reallocated

Layer Setup Requirements

WHAT MAKES EACH LAYER DECISION-READY

LAYER 6 EVALUATION — Set up BEFORE agents go live: ① Write eval rubrics — business logic & P&L constraints per agent ② Curate golden eval sets — 20–50 historical decisions with known ground truth ③ Set circuit breakers — override >15% = auto-suspend; ₹10/run cost cap Ready: ≤ 2pp accuracy drift				
LAYER 1 CAPTURE & HARMONIZE CONFIGURE: ① Connect ERP + DMS + POS via Fivetran; load 52+ weeks history into warehouse ② Configure ML anomaly checks; 98%+ quality threshold is a hard requirement ③ Build master data SKU hierarchy across all source system codes Ready when: ≥98% daily quality for 5+ consecutive days	LAYER 2 SIGNAL EXTRACTION CONFIGURE: ① Define "normal" baselines per metric / zone / category — 52-week rolling windows ② Calibrate z-score thresholds for T4 anomaly alerts (start 2.5σ; tighten over 4 weeks) ③ Map routing rules: T4 → Slack alert; T3 → decision brief queue Ready when: T4 alerts <10% false positive over 4-week calibration	LAYER 3 CAUSAL INFERENCE CONFIGURE: ① Map causal DAGs per category: trade spend → brand → consumer → purchase ② Define confounder library: seasonality, distribution, competition per domain ③ Set Lane 1/2/3 rules — which decisions auto-run vs. require human review Ready when: Lane 1 analyses match golden eval set at ≥80%	LAYER 4 INSIGHT → DECISION CONFIGURE: ① Write playbooks per decision type with guardrails, spend limits, ROI floors ② Set confidence routing matrix: >80% auto-publish; 60–80% review; <60% human ③ Define P&L guardrails: max budget shift %, minimum ROI threshold per category Ready when: briefs accepted with <5% supervisor modification rate	LAYER 5 Talent & Team ① HIRE AI EVAL LEAD Most critical new hire. Owns rubric writing, calibration tracking, drift detection, and override dashboards. ② ONBOARD SUPERVISORS 2-week training on review workflows and brief interpretation. These are your former analysts, re-roled. ③ CO-CREATE PLAYBOOKS Commercial teams co-author playbooks from Day 1. This is the single most important driver of field adoption. ④ BUILD GOLDEN EVAL SETS Domain experts curate 30 past decisions per category with known outcomes. Quality here determines agent trust.
LAYER 0 Corporate Memory	① Seed the registry: Manually enter 20–50 historical decisions per domain. This is the initial corpus agents query before any live decision runs.	② Populate causal models: Domain experts document existing knowledge — DAGs, known confounders, validated elasticity estimates per category.	③ Configure storage: Vector DB for semantic retrieval of past decisions + graph DB for SKU / outlet / category relationships. Not Notion. Not spreadsheets.	Ready when: Agents retrieve ≥3 relevant post-mortems per query type; confidence modifiers calibrated within ±5pp.

Sales Anomaly → Promo Mechanic Decision

PROTEIN+ 500G · NORTH INDIA · 02:14AM ANOMALY → 09:31AM ACTUATION

• PHASE 1 — DETECT & INVESTIGATE · FULLY AUTOMATED · 02:14AM → 02:37AM (23 MINUTES)

L2 SIGNAL EXTRACTIONT4 ANOMALY FIRED

T4 ANOMALY — Mon 02:14am

SKUProtein+ 500g

ZoneNorth India

Expected WoW+2.3%

Actual WoW-8.1%

Δ deviation-10.4pp · z = 2.8σ

→ Auto-trigger L3 investigation

→ Notify Decision Supervisor

L0 CORPORATE MEMORYQUERY + CONTEXT

MEMORY QUERY — 02:14am

"North Zone anomaly · Protein category · pricing · 2021–2024"

Found 3 post-mortems · Best match: 2023–Q2 (sim. 89%)

2023–Q2 Match

Root cause: Pricing gap +14% vs. competitor MRP avg.

Fix applied: 10% promo

Outcome: +11% sustained lift

Playbook: Pricing_Response_v3

Confidence modifier: +12%

L3 CAUSAL INFERENCE

DID · LANE 1

TreatmentNorth n=52

ControlSouth n=47

Confounders controlled:

✓ Seasonality (Q3 baseline adj.)

✓ Distribution (73%, stable)

✓ Competitor activity (neutral)

Primary Driver Identified:

Pricing gap = +15% vs. Nutrichoice

ATE = -9.8% · 95% CI [-7.2, -12.4]

Confidence: 84% → Publish to L4

• PHASE 2 — DECIDE & ACTUATE · 84% CONFIDENCE · AUTO-PUBLISHED 09:17AM · SUPERVISOR APPROVED 09:31AM

L4 INSIGHT → DECISIONAUTO-GENERATED BRIEF

DecisionPrice-off activation

SKU / ZoneProtein+ 500g, North

Mechanic12% price reduction

Duration4 weeks

Volume lift+14–18%

Incr. revenue+₹42L

Promo ROI2.4×

⚠ Caveat: monitor competitor response wk2

CONFIDENCE: 84% · AUTO-PUBLISH

L6 EVALUATION GATEALL CHECKS PASSED

Confidence84% ✓ (>80%)

Routing tierAUTO-PUBLISH

Golden eval match87% ✓

Override rate 7d3.4% ✓

Cost / run₹6.80 ✓

P&L risk checkWithin limits ✓

Causal model used: Pricing_v7

Playbook: Pricing_Response_v3

APPROVED FOR DELIVERY

L5 SUPERVISORAPPROVED ✓ 14 MIN

09:17amBrief delivered

09:31am★ Approved (14 min)

API write-backs executed:

✓ ERP: price updated (-12%)

✓ DMS: distributor briefed

✓ CRM: trade plan logged

L0 memory updated:

Decision logged · post-mortem due +28d

Anomaly → Actuation: 7h 17m

02:14amT4 Anomaly

23 min · automated

02:37amBrief Ready

queued overnight

09:17amSupervisor Notified

14 min review

09:31amApproved + Actuated

Q3 Trade Spend Reallocation at Scale

450 OUTLETS · ₹18.4 CR BUDGET · NORTH + WEST INDIA · 50,000+ MICRO-ANALYSES

INPUT DATA PROFILE — L1 + L2

Scope

North + West India

Outlets analyzed

450

Priority SKUs

8

Mechanics tested

6 types

Budget pool

₹18.4 Cr

Data rows (L1)

3.2M · 99.4% ✓

PLATFORM PROCESSING LOG

L1

DMS + POS + ERP ingested
11:30pm overnight batch · 3.2M rows

L2

T1 + T2 + T4 signals extracted
23 forward-buy flags · 167 zero-lift outlets

L0

3 playbooks queried from registry
Trade_Spend_v3 · Promo_Mech_v4 · Payout_v2

L3

450 cohort analyses run overnight
Mechanic elasticity per segment measured

L4

Reallocation brief auto-generated
Confidence: 77% → Review Required

L6

Eval gate: 77% → routed to supervisor
Golden eval match: 82% ✓ · Cost: ₹8.20/run

L5

Supervisor reviewed + modified + approved
L0 playbook updated → v4.2

L3 CAUSAL FINDING — AVG. NET LIFT PER MECHANIC (ALL 450 OUTLETS)

BOGO

+5.2%

High cannibalization
Subsidizes fwd. buying

Fixed Display

+3.4%

Location-dependent
Low scalability

20% Off-Price

+12.1%

Sustained ROI
Low post-promo dip

Perf-Linked

+17.8%

Best net lift
Eliminates fwd. buy

L4 AGENT RECOMMENDATION + L5 SUPERVISOR MODIFICATION

APPROVED WITH MODIFICATION

Mechanic	Current	Agent Rec.	Supervisor Final	Rationale
BOGO	₹6.2 Cr	₹2.0 Cr ↓	₹3.1 Cr ↓	"Keep for 15 premium outlets — key distributor relationships"
20% Off-Price	₹4.1 Cr	₹6.2 Cr ↑	₹6.2 Cr ↑	+12.1% net lift · high causal confidence
Perf-Linked	₹2.3 Cr	₹4.4 Cr ↑	₹4.4 Cr ↑	+17.8% net lift · eliminates forward-buying
Fixed Display	₹3.1 Cr	₹3.1 Cr →	₹3.1 Cr →	Unchanged · location strategy
Other	₹2.7 Cr	₹2.7 Cr →	₹2.7 Cr →	Unchanged

PROJECTED IMPACT

Incremental off-take

+₹3.1 Cr

Avg. net lift improvement

+7.2%

Fwd-buy waste eliminated

₹4.8 Cr

L0 MEMORY WRITE — PLAYBOOK V4.2

✓ Trade_Spend playbook updated to v4.2

✓ New rule: Premium outlet BOGO exception

✓ 450 outlet decisions + supervisor edit logged

→ Post-mortem scheduled: +6 weeks